Groups

$(G, *)$ is a group if it satisfies

(i) Closure property

$$a*b \in G \quad \forall\ a,b \in G$$

(ii) Associative property

$$(a*b)*c = a*(b*c) \quad \forall\ a,b,c \in G$$

(iii) Identity : $\exists e \in G$ s.t

$$a*e = e*a = a \quad \forall\ a \in G$$

(iv) Inverse : $\forall a \in G\ \exists\ b \in G$ s.t

$$a*b = e = b*a$$

identity element

(v) Commutative : $a*b = b*a \quad \forall\ a,b \in G$

A group which also satisfies commutative property is known as "Abelian Group"

Q. Show that set of all positive rational numbers form an Abelian group under the composition defines by

$$a*b = \frac{ab}{2}$$

1. Closure

$$a \in Q^+$$
$$b \in Q^+$$

$$\therefore\ a*b = \frac{ab}{2} \in Q^+$$

Hence, $a*b \in Q^+$

2. Associative

$= (a * b) * c$

$= \left(\frac{ab}{2}\right) * c$

$= \frac{\left(\frac{ab}{2}\right)c}{2} = \frac{abc}{4}$

And $a * (b * c)$

$= a * \left(\frac{bc}{2}\right)$

$= \frac{a\left(\frac{bc}{2}\right)}{2} = \frac{abc}{4}$

$\therefore \quad (a * b) * c = a * (b * c)$

3. Identity ; Let e be the identity element

$\Rightarrow \quad a * e = a$

$\therefore \quad \frac{ae}{2} = a \qquad \Rightarrow \quad e * a = a$

$\qquad\qquad\qquad\qquad \frac{2 \times a}{2} = a$

$e = 2 \qquad\qquad\qquad a = a$

And, as $e \in Q^+$

Hence, $(Q^+, *)$ have an identity element.

4. Inverse ;

$\Rightarrow \quad a * b = e$

$\therefore \quad \frac{ab}{2} = 2$

$b = 4/a$

As, $a \in Q^+$

$\Rightarrow b \in Q^+$

Hence, inverse element $\Rightarrow 4/a$.

5. Commutative

$$a * b = \frac{ab}{2}$$

and, $b * a = \frac{ba}{2} = \frac{ab}{2}$

$\Rightarrow \quad a * b = b * a$

Hence, $(Q^+, *)$ satisfies all 5 properties.

$\therefore$ It is an Abelian group.

Properties of Group

1. Identity element in a group is unique

Proof: Let e and e' be 2 distinct identity elements in a group.

$$e, e' \in G$$

$e * e' = e'$ (where e is identity)

$[\because a * e = a \ \forall \ a \in G]$

$e * e' = e$ (where e' is identity)

$\Rightarrow \quad e' = e * e' = e$

Hence, identity element is unique.

2. Inverse of an element in a group is unique.

Proof: Let $a \in G$ have 2 inverse, say b and c.

$\Rightarrow \quad a * b = e = b * a$ —— (1)

$a * c = e = c * a$ —— (2)

$b = b * e = b * (a * c)$ From (2)

$= (b * a) * c$ Associative Property

$= e * c$ From (1)

$= c$

3. Prove that $(ab)^{-1} = b^{-1}a^{-1} \;\; \forall\, a,b \in G$

$$\left.\begin{array}{ll} \stackrel{?}{=} & (ab)(b^{-1}a^{-1}) = e \\ \text{and} & (b^{-1}a^{-1})(ab) = e \end{array}\right\} \text{ To prove}$$

$$\Rightarrow (ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1}$$
$$= a\,e\,a^{-1}$$
$$= a\,a^{-1}$$
$$= e$$

Similarly,

$$(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b$$
$$= b^{-1}.e.b$$
$$= b^{-1}b$$
$$= e$$

4. Prove that $(a^{-1})^{-1} = a \quad \forall\; a \in G$

$$\stackrel{?}{=} \quad (a^{-1})a = e$$
$$\text{and} \quad a(a^{-1}) = e$$

$\Rightarrow$ $(a^{-1})a = e$ and similarly $a(a^{-1}) = e$.

Cancellation Law

If $a, b, c, \in G$, then

(i) $ab = ac \Rightarrow b = c$ (Left cancellation Law)

(ii) $ba = ca \Rightarrow b = c$ (Right cancellation Law)

Proof (i) $ab = ac$

Premultiplying both sides by a^{-1}

$$a^{-1}(ab) = a^{-1}(ac)$$
$$(a^{-1}a)b = (a^{-1}a)c \quad \text{Associative property}$$
$$e\,b = e\,c$$
$$b = c$$

Q. Consider $(\mathbb{Z}, *)$ where $a*b = a+b+1 \ \forall\ a,b \in \mathbb{Z}$
Check if this forms a group?

1. Closure

$a \in \mathbb{Z}$,
$b \in \mathbb{Z}$

$\therefore (a*b) = a+b+1 \in \mathbb{Z}$

2. Associative

$$\underbrace{(a*b)}_{g}*c = g*c$$
$$= g+c+1$$
$$= (a*b)+c+1$$
$$= a+b+1+c+1$$
$$= a + \underbrace{(b+c+1)}_{h = b*c}+1$$
$$= a + h + 1$$
$$= a*h$$
$$= a*(b*c)$$

3. Identity

$$a * e = a$$
$$a + e + 1 = a$$
$$e = -1$$

4. Inverse

$a * a^{-1} = e$ and $a^{-1} * a = e$

$a + a^{-1} + 1 = -1$ and $a^{-1} + a + 1 = -1$

$$a^{-1} = -2 - a \ \forall\ a \in \mathbb{Z}$$

Hence, $(\mathbb{Z}, *)$ is a group.

Finite Groups : $(G, *) \rightarrow G < \infty$

Show that fourth root of unity form a group

$\rightarrow$ $G = \{1, -1, i, -i\}$ $\quad * \rightarrow .$

$*$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

All the elements of the composition table $\in G$.
Hence, closure property holds

Associativity: Associativity holds for complex numbers.

Identity: $a \cdot e = a$
$e = 1$

Inverse: $1^{-1} = 1$, $(-1)^{-1} = -1$

$i^{-1} = -i$, $(-i)^{-1} = i$

Hence, inverse also holds.

Q. Prove that $G = \{0, 1, 2, 3, 4, 5\}$ is a finite Abelian group w.r.t addition modulo 6 $(+_6)$

To prove: $(G, +_6)$ is an Abelian group

$a +_6 b \rightarrow$ remainder when $a+6$ divided by 6.

Composition Table

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(i) Closure : Closure property holds from table.

(ii) Associativity : $(3 +_6 4) +_6 5$

$= 1 +_6 5 = 0$

and

$3 +_6 (4 +_6 5)$

$= 3 +_6 3 = 0$

(iii) Identity : $e = 0$

(iv) Inverse : $0^{-1} = 0$, $1^{-1} = 5$, $2^{-1} = 4$, $3^{-1} = 3$, $4^{-1} = 2$, $5^{-1} = 1$

(v) Commutative : $(3 +_6 4) = 1$

and $(4 +_6 3) = 1$

Hence, $(G, +_6)$ is an Abelian group.

Q. Prove that $G = \{1, 2, 3, 4, 5, 6\}$ is a finite Abelian group w.r.t multiplication modulo 7 $(\times_7)$.

To prove : $(G, \times_7)$ is an Abelian group.

$a \times_7 b \rightarrow$ Remainder when $a \cdot b$ is divided by 7

Composition Table :

$\times_7$	0	1	2	3	4	5	6
0							
1		1	2	3	4	5	6
2		2	4	6	1	3	5
3		3	6	2	5	1	4
4		4	1	5	2	6	3
5		5	3	1	6	4	2
6		6	5	4	3	2	1

Associativity: $2 \times_7 (3 \times_7 4) = 2 \times_7 5 = 3$
$(2 \times_7 3) \times_7 4 = 6 \times_7 4 = 3$

Identity: $e = 1$

Inverse: $1^{-1} = 1$, $2^{-1} = 4$, $3^{-1} = 5$, $4^{-1} = 2$, $5^{-1} = 3$, $6^{-1} = 6$

Commutative: $(2 \times_7 3) = 6$ and
$(3 \times_7 2) = 6$

Hence, it is an Abelian group.

Subgroup

$H \subseteq G$ $(G, *) \to$ group then if $(H, *) \to$ group

$(\mathbb{Z}, +) \to$ group
$(\mathbb{Z}_m, +) \to$ subgroup of $(\mathbb{Z}, +)$
$\mathbb{Z}_m = \{\ldots, -2m, -m, 0, m, 2m, \ldots\}$

Theorem: A necessary and sufficient condition for a non-empty subset H of G to be a subgroup is that $a, b \in H$ implies $a * b^{-1} \in H$ where b^{-1} is the inverse of b in G.

Q. $G = \mathbb{Z}$, $(\mathbb{Z}, +)$ is a group
Let $H = \mathbb{Z}_m = \{\ldots, -2m, -m, 0, m, 2m, \ldots\}$
Clearly $H \subset G$

T.P.T: $(H, +)$ is a subgroup of G
$\Rightarrow$ Let $a, b \in H$
$a = nm$, $b = sm$, $n, s \in \mathbb{Z}$
$b^{-1} = -sm \in H$, $-s \in \mathbb{Z}$

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Q $G=\{1,2,3,4,5\}$, $(G, \times_6)$

$\times_6 \rightarrow$ multiplication modulo 6

$\Rightarrow$

$\times_6$	1	2	3	4	5	6
1	1	2	3	4	5	0
2	2	4	0	2	4	0
3	3	0	3	0	3	0
4	4	2	0	4	2	0
5	5	4	3	2	1	0
6	0	0	0	0	0	0

$\rightarrow$ Here, we are having two identity elements 1&6 which are not possible

It'll not be a group since it is a composite number.

Order of an element in a group

Let $a \in G$, then order of an element a is n if n is least positive integer if it exists such that $a^n = e$

Ex- $G=\{1,-1,i,-i\}$ $(G, \times)$

$e=1$

$(1)^1 = 1$ $\quad (O(1)=1)$

$(-1)^2 = 1$ $\quad (O(-1)=2$

$(+i)^4 = 1$ $\quad O(i)=4$

$(-i)^4 = 1$ $\quad O(-i)=4$

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Q $G = \{0, 1, 2, 3, 4, 5\}$ $(G, +_6)$

$\Rightarrow$ $e = 0$

~~i=1~~ $1 = 1$

$1^2 = 1 +_6 1 = 2$

$1^3 = 2 +_6 1 = 3$

$1^4 = 3 +_6 1 = 4$

$1^5 = 4 +_6 1 = 5$

$1^6 = 5 +_6 1 = 0$

Hence, $O(1) = 6$

$2^1 = 2$

$2^2 = 2 +_6 2 = 4$

$2^3 = 4 +_6 2 = 0$ Hence, $O(2) = 3$

$3^1 = 3$

$3^2 = 3 +_6 3 = 0$ Hence, $O(3) = 2$

$4^1 = 4$

$4^2 = 4 +_6 4 = 2$ Hence, $O(4) = 3$

$4^3 = 2 +_6 4 = 0$

$5^1 = 5$

$5^2 = 5 +_6 5 = 4$

$5^3 = 4 +_6 5 = 3$

$5^4 = 3 +_6 5 = 2$

$5^5 = 2 +_6 5 = 1$ Hence, $O(5) = 6$

$5^6 = 1 +_6 5 = 0$

$O(1) = O(5) = 6$

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Q. Proof that if G is an abelian group, then $\forall\, a, b \in G$ & all integers n, $(ab)^n = a^n \cdot b^n$.

$\Rightarrow$ Case I: $n=0$, $(ab)^0 = 1 = a^0 b^0$

Case II: $n>0$, for $n=1$, $(ab)^1 = ab$

Assume for $n=k$ (i.e. $(ab)^k = a^k b^k$) — (1)

Consider $(ab)^{k+1} = (ab)^k (ab)$

$= (a^k \cdot b^k)(ab)$ From (1)

$= a^k \cdot (b^k a) b$ Assoc. Prop.

$= a^k \cdot (ab^k) b$ ($\because G$ is abelian)

$= a^{k+1} \cdot b^{k+1}$

Case III: $n<0$, let $n=-m$, $m>0$ integer

$(ab)^n = (ab)^{-m}$

$= [(ab)^m]^{-1}$

$= [b^m \cdot a^m]^{-1}$

$= a^{-m} \cdot b^{-m}$

$= a^n \cdot b^n$

Q. Proof that $a^2 = e$. $\forall\, a \in G$ when G is Abelian.

or

Show that if every element of a group G is its own inverse, then G is Abelian.

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$\Rightarrow$ $a^2 = e \Rightarrow a^{-1} = a$ —— (1)

$b^2 = e \Rightarrow b^{-1} = b$ —— (2)

To prove that : $ab = ba \;\forall\; a, b \in G$

$$a, b \in G \Rightarrow ab \in G$$
$$\Rightarrow (ab)^2 = e$$
$$\Rightarrow (ab)^{-1} = ab$$
$$\Rightarrow b^{-1}a^{-1} = ab$$

$ba = ab$ (From (1) & (2))

$\Rightarrow$ G is abelian.

Q. Show that if a, b are any two elements of a group G, then $(ab)^2 = a^2b^2$ iff G is Abelian.

$\Rightarrow$

$$(ab)(ab) = a \cdot a \cdot b \cdot b$$
$$a \cdot (ba) b = a \cdot a \cdot b \cdot b$$

Using left cancellation law,

$$(ba) \cdot b = a \cdot b \cdot b$$

Using right cancellation law,

$ba = ab$ Hence, commutative & Abelian

Subgroup : A subset H of G $(H \subset G)$ is said to be a subgroup if $(H, *)$ is a group in itself.

Associativity: $2 \times_7 (3 \times_7 4) = 2 \times_7 5 = 3$
$(2 \times_7 3) \times_7 4 = 6 \times_7 4 = 3$

Identity: $e = 1$

Inverse: $1^{-1} = 1,\ 2^{-1} = 4,\ 3^{-1} = 5,\ 4^{-1} = 2,\ 5^{-1} = 3,\ 6^{-1} = 6$

Commutative: $(2 \times_7 3) = 6$ and
$(3 \times_7 2) = 6$

Hence, it is an Abelian group.

Subgroup

$H \subseteq G$ $(G, *) \to$ group then if $(H, *) \to$ group

$(\mathbb{Z}, +) \to$ group
$(\mathbb{Z}_m, +) \to$ subgroup of $(\mathbb{Z}, +)$
$\mathbb{Z}_m = \{ \dots, -2m, -m, 0, m, 2m, \dots \}$

Theorem: A necessary and sufficient condition for a non-empty subset H of G to be a subgroup is that $a, b \in H$ implies $a * b^{-1} \in H$ where b^{-1} is the inverse of b in G.

Q. $G = \mathbb{Z}$, $(\mathbb{Z}, +)$ is a group
Let $H = \mathbb{Z}_m = \{ \dots, -2m, -m, 0, m, 2m, \dots \}$
Clearly $H \subset G$

T.P.T: $(H, +)$ is a subgroup of G
$\Rightarrow$ Let $a, b \in H$
$a = nm$, $b = sm$, $n, s \in \mathbb{Z}$
$b^{-1} = -sm \in H$, $-s \in \mathbb{Z}$

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Let $a, b^{-1} \in H$, $a * b^{-1} = nm + (-sm)$
$= (n-s)m$, $n-s \in \mathbb{Z}$

$\Rightarrow (H, +)$ is a subgroup of $(G, +)$.

Q. Prove that H is a subgroup of group, $(\mathbb{R}, \times)$.
$H = \{a + b\sqrt{2} : a, b \in \mathbb{R}\}$ where $\mathbb{R}_0 = \mathbb{R} - \{0\}$.

$\Rightarrow$ Let $A = a + b\sqrt{2}$ and $C = c + d\sqrt{2} \in H$, where
$a, b, c, d \in R_0$

$$C^{-1} = \frac{1}{c + d\sqrt{2}} \times \frac{c - d\sqrt{2}}{c - d\sqrt{2}}$$

$$= \frac{c - d\sqrt{2}}{c^2 - 2d^2} = c_1 + d_1\sqrt{2} \in H$$

where $c_1 = \dfrac{c}{c^2 - 2d^2}$ and $d_1 = \dfrac{-d}{c^2 - 2d^2} \in \mathbb{R}$

$A, C^{-1} \in H \Rightarrow AC^{-1} = (a + b\sqrt{2})(c_1 + d_1\sqrt{2})$
$= ac_1 + ad_1\sqrt{2} + bc_1 + 2bd_1$
$= a_1 + b_1\sqrt{2}$

where $a_1 = ac_1 + 2d_1$
$b_1 = bc_1 + ad_1$

Q. If G is a set of ordered pair (a, b) where $a \neq 0$ and binary operation in G is defined as

$$(a, b) * (c, d) = (ac, bc + d)$$

Show that $(G, *)$ is a group. Show also that subset H of all elements of G which are of this non-abelian form it is a subgroup of G.

$\Rightarrow G = \{(a,b) : a \neq 0,\ a, b \in R\}$

T.P.T: 1. $(G, *)$ is a non-abelian
2. $H = \{a, b : b \in R\}$. $(H, *)$ is a subgroup of $(G, *)$

$\Rightarrow (a,b) \in G$ and $(c,d) \in G \Rightarrow a \neq 0, c \neq 0$

Associativity: $[(a,b) * (c,d)] * (k,m) =$

$= (ac,\ bc+d) * (k,m)$

$= (ack,\ (bc+d)k + m)$

$= [a(ck),\ bck + (dk+m)]$

$= [(a,b)] * [ck,\ dk+m]$

$= [(a,b) * ((c,d) * (k,m))]$

Identity

Let (e,f) be the identity element in n,

$(a,b) * (e,f) = (a,b)$

$\therefore (ae + be + f) = (a,b)$

$\Rightarrow ae = a$

$e = 1$

and $be + f = b$

$\Rightarrow f = 0$

$\Rightarrow (e,f) = (1,0)$

Inverse,

$(a,b) * (c,d) = (1,0)$

$\therefore (ac,\ bc+d) = (1,0)$

$c = 1/a$ and $d = -bc = -b/a$

2. $H = \{(1, b) : b \in \mathbb{R}\}$, $(H, *)$ is a subgroup of $(G, +)$

Let $\alpha, \beta = H \Rightarrow \alpha = (1, b_1)$, $\beta = (1, b_2)$, $b_1, b_2 \in \mathbb{R}$

$$\beta^{-1} = (1, b_2)^{-1} = \left(1, -\frac{1}{b_2}\right) \in G$$

$$\alpha * \beta^{-1} = (1, b_1) * \left(1, -\frac{1}{b_2}\right)$$

$$= \left(1, b_1 - \frac{1}{b_2}\right) \in H$$

$\Rightarrow$ $(H, *)$ is a subgroup.

Q. $(G, *) \to$ group

$(H_1, *)$ and $(H_2, *) \to$ subgroup

$(H_1 \cap H_2, *)$ and $(H_1 \cup H_2, *) \to$?

$(H_1 \cap H_2, *)$ $\downarrow$ Always subgroup of G

$(H_1 \cup H_2, *)$ $\downarrow$ Not necessarily subgroup of G.

▫ Cyclic Group

A group G along with binary operation $*$ is said to be cyclic $\exists\, a \in G$ s.t every element of G can be written in some power of a.

$$G = \{a^m : m \in \mathbb{Z}\}$$

and a is called as generator of G

$$G = \langle a \rangle$$

→ Theorem: Every cyclic group is Abelian.

Let G be a cyclic group.

T.P.T : G is Abelian.

Let $x, y \in G$

Let $G = \langle a \rangle$

$\Rightarrow \exists\ p, q \in \mathbb{Z}$ s.t

$x = a^p$

and $y = a^q$

$\therefore\ xy = a^p . a^q = a^{p+q} = a^{q+p} = a^q . a^p = yx$

Hence, G is Abelian.

Theorem: The order of cyclic group is same as the order of it's generator.

T.P.T: Let $G = \langle a \rangle$ and $O(a) = n \Rightarrow n$ is the smallest +ve integer s.t $a^n = e$

$\# G = n$

Let $G = \{a, a^2, a^3, \dots, a^n\}$

Case 1: If all elements in G are distinct then

$\#(G) = n = O(a)$.

Case 2: Suppose 2 elements are same i.e $a^p = a^q$, $p = q$

$\Rightarrow a^{p-q} = e \quad \# \quad a \equiv p-q < n$

a^p and a^q are distinct

$\Rightarrow$ All elements in G are distinct

$\#(G) = O(a)$

Q. Prove that order of an element of a group is same as that of it's inverse.

$a \in G$

$O(a) = n$

T.P.T $\quad O(a^{-1}) = n$

Q. Every group of order 4 is Abelian.

$G = \{a, b, c, e\}$ $\quad \#(G) = 4$.

T.P.T $(G, *)$ is an Abelian group.

Case 1: $a^{-1} = a, \; b^{-1} = b, \; c^{-1} = c$

$\Rightarrow$ G is an Abelian group

Case 2: $a^{-1} = b \; , \; b^{-1} = a \; , \; c^{-1} = c$

*	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

From composition table;

$c * a = a * c = b$

$\Rightarrow$ G is an Abelian group.

□ If a cyclic group G is generated by a of order n.
Then a^m is generated of G iff
$\gcd(m, n) = 1$.

$G = \langle a \rangle \quad , \quad O(a) = n$

a^m iff $\gcd(m, n) = 1$

$G = \langle a^m \rangle$

Q. $G = \{0, 1, 2, 3, 4, 5\}$ is a finite Abelian group. What would be it's order? Find all generators?
Group is Abelian w.r.t addition modulo 6 ($+_6$)

$\cancel{P} =$

$\therefore$
$1^1 = 1$
$1^2 = 1 +_6 1 = 2$
$1^3 = 1^2 +_6 1 = 3$
$1^4 = 1^3 +_6 1 = 4$
$1^5 = 1^4 +_6 1 = 5$
$1^6 = 1^5 +_6 1 = 0$

$\therefore$ Order of $G = 6$
One of the generator of $G = \langle 1 \rangle$

$\therefore$ $\gcd(m, 6) = 1$
$\Rightarrow m = 5$

and $a^m = 1^5 = 5$ Ans

$\therefore$ G have two generators = $\langle 1 \rangle$ and $\langle 5 \rangle$

Q. $G = \{1, 2, 3, 4, 5, 6\}$ is a finite Abelian group w.r.t multiplication modula 7 (X_7).
What would be it's order and generators?

$1^1 = 1$
$1^2 = 1 X_7 1 = 1$
$1^3 = 1^2 X_7 1 = 1$

Repeating, hence 1 cannot be generator X

$2^1 = 2$
$2^2 = 4 \% 7 = 4 \Rightarrow 2 X_7 2 = 4$
$2^3 = 8 \% 7 = 1 \Rightarrow 2^2 X_7 2 = 1$
$2^4 = 2^3 X_7 2 = 2$
$2^5 = 2^4 X_7 2 = 4$

Again repeating
2, 4, 1, 2, 4...

2 cannot be generator X

$\therefore$ $3^1 = 3$

$3^2 = 3 \times_7 3 = 2$

$3^3 = 3^2 \times_7 3 = 6$

$3^4 = 3^3 \times_7 3 = 4$

$3^5 = 3^4 \times_7 3 = 5$

$3^6 = 3^5 \times_7 3 = 1$

$\therefore$ 3 is a generator of G
Hence, order of $G = 6$

$\therefore$ $\gcd(m, 6) = 1$

$m = 5$

$\Rightarrow$ $G = \langle 3^5 \rangle = \langle 5 \rangle$ Ans

Hence G have two generators : $\langle 3 \rangle$ and $\langle 5 \rangle$ Ans.

Q. Find all generator of cyclic group of order 8

$\therefore$ Let $G = \langle a \rangle$

Now $\gcd(m, 8) = 1$

$m = 3, 5, 7$

$\therefore$ $G = \langle a^3 \rangle, \langle a^5 \rangle, \langle a^7 \rangle$ Ans.

Q. How many generators are there cyclic group of order 10.

$\therefore$ Let $G = \langle a \rangle$

Now, $\gcd(m, 10) = 1$

$m = 3, 7, 9$

$\therefore$ $G = \langle a^3 \rangle, \langle a^7 \rangle, \langle a^9 \rangle.$

Hence 3 generators.